

Tuhin Mandal

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GitHub | LinkedIn | Portfolio | LeetCode

PROFESSIONAL SUMMARY

Software Engineer with a B.Tech in CSE and a strong foundation in modern backend and frontend architectures. Proficient in building scalable APIs using **FastAPI** and **Node.js**, with a focus on asynchronous programming and data validation. Advanced problem-solver with **215+ LeetCode** problems solved, demonstrating expertise in algorithmic efficiency and system logic.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), JavaScript (ES6+), C++, HTML5, CSS3, SQL
- **Frameworks/Libraries:** FastAPI, Next.js, React, Node.js, Express.js, Pydantic, Zod
- **Databases & Tools:** PostgreSQL, SQLite, RESTful APIs, Git, GitHub, Linux, Docker
- **Modern Tooling:** uv (Python Package Manager), Vite, Vercel

PROJECTS

LeetCode Tracker API | *Python, FastAPI, Pydantic, SQLite* 2025

- Engineered a high-performance backend for tracking algorithmic progress using **FastAPI** and **asynchronous** request handling.
- Implemented robust data validation and serialization using **Pydantic**, ensuring type safety across the application lifecycle.
- Optimized development workflow by using **uv** for deterministic dependency management, ensuring 100% reproducible environments.

Developer Portfolio & Showcase | *Next.js, React, Tailwind CSS* 2025

- Developed a responsive, high-performance portfolio website using **Next.js** to showcase technical projects and skills.
- Utilized modern component architecture and server-side rendering (SSR) for optimized SEO and fast load times.
- Deployed via Vercel with continuous integration to ensure immediate updates upon code changes.

KPA Forms API | *Node.js, Express, PostgreSQL, Zod* 2024

- Built a secure RESTful API for managing form submissions with **Zod** schema validation and parameterized SQL queries.
- Containerized the database environment using **Docker** to streamline local development and ensure deployment consistency.

EDUCATION

B.Tech in Computer Science & Engineering 2020 – 2024

Chaibasa Engineering College, Jharkhand

CGPA: 8.11

TECHNICAL ACHIEVEMENTS

- **LeetCode Proficiency:** Solved **215+** algorithmic problems, consistently applying optimal time and space complexity ($O(n)$, $O(n \log n)$).
- **Open Source / Tooling:** Early adopter of **uv** (by Astral) for Python project management, achieving significantly faster install times and stable builds.